

Food Delivery Time Prediction

Objective

The goal is to predict food delivery times based on customer location, restaurant location, weather, traffic, and other factors. This involves both **data preprocessing** and building predictive models using **linear regression** and **logistic regression**.

Phase 1: Data Collection and Exploratory Data Analysis (EDA)

Step 1 - Data Import and Preprocessing

1. Dataset

Load the dataset ([Food_Delivery_Time_Prediction.csv](#)).

2. Handle Missing Values

Check for any missing or inconsistent values in columns such as Distance, Delivery_Time, etc. and decide how to handle them, either through imputation or deletion.

3. Data Transformation

- **Encode Categorical Variables:** Use one-hot encoding or label encoding for variables like Weather Conditions, Traffic Conditions, Vehicle Type.
- **Normalize/Standardize Numeric Columns:** Normalize or standardize continuous features like Distance, Delivery_sTime, and Order_Cost for consistency.

Step 2 - Exploratory Data Analysis (EDA)

1. Descriptive Statistics

Calculate the basic statistics for numerical features such as mean, median, mode, and variance.

2. Correlation Analysis

Visualize correlations between features and the target variable (Delivery_Time) to identify the most relevant predictors.

3. Outlier Detection

Detect outliers in numerical features using boxplots and handle them appropriately.

Step 3 - Feature Engineering

1. Distance Calculation

If the dataset doesn't contain an actual distance metric, calculate the distance between the customer and restaurant using latitudes and longitudes (Haversine formula).

2. Time-Based Features

Create new features related to the time of day, such as Rush Hour vs Non-Rush Hour, to improve predictions.

Phase 2: Predictive Modeling

Step 4 - Linear Regression Model

1. Train-Test Split

Split the dataset into training and testing sets (e.g., 80/20 split).

2. **Model Building**

Use **Linear Regression** to predict the Delivery Time based on features like Distance, Traffic_Conditions, and Order_Priority.

3. **Evaluation Metrics**

Evaluate the model using:

- **Mean Squared Error (MSE)**
- **R-squared (R^2)**
- **Mean Absolute Error (MAE)**

Step 5 - Logistic Regression Model (for Categorization)

1. **Model Objective**

Classify deliveries as "Fast" or "Delayed" based on binary features such as Traffic, Weather, Delivery_Person_Experience, etc.

2. **Model Implementation**

Use **Logistic Regression** to predict the delivery status.

3. **Evaluation Metrics**

Evaluate using metrics such as:

- **Accuracy**
- **Precision**
- **Recall**
- **F1-score**
- **Confusion Matrix**

Phase 3: Reporting and Insights

Step 6 - Model Evaluation and Comparison

- Compare the **Linear Regression** and **Logistic Regression** models based on their performance (e.g., accuracy, confusion matrix).
- Visualize the results using **confusion matrices** and **ROC curves**.

Step 7 - Actionable Insights

- Based on model predictions, suggest operational improvements such as:
 - Optimizing delivery routes.
 - Adjusting staffing during high-traffic periods.
 - Providing better training to delivery staff

Final Deliverables

1. **Jupyter Notebook (.ipynb):**

- Complete code for data preprocessing, model training, and evaluation.

2. **Data Visualizations:**

- Visual representations such as **scatter plots**, **pair plots**, **confusion matrices**, and **ROC curves** to interpret the results.

3. **Final Report:**

- A detailed summary of the project, including:
 - Description of the dataset and preprocessing steps.
 - Model evaluation and comparisons.
 - Actionable insights and recommendations for optimization.